

# **Bayesian Predictive Modeling for Metabolic Disease Risk in Cattle: Translational Framework Integrating Serum Biomarkers and Behavioral Indicators**

**David Iyileh**

Warren Wilson College Environmental Studies Department

Elizabeth Benavides, PhD; Langdon Martin, PhD; Christopher Potvin, PhD

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## Abstract

Metabolic disorders in cattle, particularly milk fever (hypocalcemia) and grass tetany (hypomagnesemia), remain persistent challenges for herd health and economic sustainability. Traditional diagnostics rely on threshold-based mineral cutoffs and visual symptom detection, which fail to account for individual variation in physiology, behavior, and environment.

This study integrates serum mineral concentrations (Ca, Mg, K) with behavioral indicators (chute score, exit velocity, hair whorl position) and performance metrics (body weight, reproductive efficiency, body condition score) to develop Bayesian logistic regression models for individualized risk prediction. Blood samples from 39 cattle at the Warren Wilson College Farm were collected via coccygeal venipuncture, processed to serum, and analyzed using inductively coupled plasma optical emission spectroscopy (ICP-OES) for calcium and magnesium, and atomic absorption spectroscopy (AAS) for potassium. Calibration curves were constructed from LabChem standards, and acid digestion protocols ensured accurate mineral quantification.

Risk categories were assigned based on established veterinary clinical thresholds for subclinical disease: milk fever risk ( $\text{Ca} < 8.0 \text{ mg/dL}$ ) and grass tetany risk ( $\text{Mg} < 1.6 \text{ mmol/L}$  or  $\text{K:Mg} > 3.0$ ). Models were evaluated using ROC and precision-recall curves, confusion matrices, and posterior predictive checks. Results demonstrated strong discrimination for milk fever (PR-AUC = 0.9832, accuracy = 89.7%) and moderate performance for grass tetany (PR-AUC = 0.6788, accuracy = 84.6%). Posterior estimates confirmed physiologically plausible associations: low Ca strongly predicted milk fever risk, while low Mg and elevated K:Mg ratios aligned with grass tetany physiology.

By bridging spectroscopy with behavioral predictors, this framework advances proactive herd health diagnostics, supports reproducible workflows, and contributes translational relevance for human metabolic health, where mineral imbalance and stress physiology interact similarly.

**Keywords:** Bayesian modeling; calcium; cattle; magnesium; metabolic disorders; potassium

## Introduction

Metabolic diseases such as milk fever (hypocalcemia) and grass tetany (hypomagnesemia) remain persistent barriers to sustainable cattle production, with serious consequences for animal welfare, productivity, and farm economics. These disorders often emerge abruptly, leaving limited opportunity for effective treatment. Traditional diagnostic practices rely on generalized herd-level thresholds and visual symptom detection (Goff, 2008; Martens & Schweigel, 2000), approaches that overlook individual variation and fail to provide proactive, data-driven risk assessment. As a result, many cases of subclinical disease go undetected until they progress to more severe stages, reducing reproductive efficiency, milk yield, and overall herd performance.

Across the cattle industry, farms and researchers have adopted a variety of strategies to manage these conditions. Large commercial dairies often rely on routine blood sampling, dietary supplementation, and mineral boluses to reduce risk, while some operations use precision feeding systems to balance calcium, magnesium, and potassium intake. Researchers have emphasized the importance of monitoring mineral ratios, particularly the antagonistic relationship between potassium and magnesium in forage, as a predictive tool for grass tetany. However, these approaches are often resource-intensive, require frequent sampling, or depend on infrastructure not available to smaller, pasture-based farms.

At Warren Wilson College Farm, cattle are maintained under rotational grazing systems and are on pasture year-round. Blood collection opportunities are limited to routine herd work through the chute system, typically only a few times per year. This makes it especially important to maximize the diagnostic value of each sample collected. Following Hurricane Helene, which caused catastrophic flooding in the Swannanoa Valley and disrupted pasture composition, monitoring herd mineral status has become even more critical. Shifts in forage quality and availability can alter mineral intake, increasing the risk of metabolic disorders. Unlike larger operations, the WWC Farm must balance herd health monitoring with limited resources, making reproducible, efficient diagnostic frameworks essential.

The present study focuses exclusively on serum samples from cattle, quantified for calcium (Ca), magnesium (Mg), and potassium (K) concentrations. Calcium and magnesium were measured using inductively coupled plasma optical emission spectroscopy (ICP-OES), while potassium was quantified separately using atomic absorption spectroscopy (AAS) due to instrument detection limitations. This dual-instrument approach ensures accurate mineral profiling and establishes serum-based monitoring as a reproducible diagnostic framework for herd health.

Beyond mineral quantification, this research integrates behavioral indicators (chute score, exit velocity, hair whorl position) and performance metrics (body weight, reproductive efficiency, body condition score) into Bayesian logistic regression models. By combining mineral biomarkers with behavioral and performance data, the framework advances individualized, probabilistic risk prediction for metabolic disorders. This approach moves beyond binary thresholds to continuous risk scores with quantified uncertainty, offering a more nuanced understanding of disease risk.

For Warren Wilson College Farm, the implications are direct and practical. By leveraging serum samples collected during routine herd work, this study provides a method to proactively identify animals at higher risk of metabolic disorders before clinical symptoms appear. This enables caretakers to make informed management decisions, such as targeted supplementation or closer monitoring of vulnerable individuals, without requiring costly or frequent sampling. More broadly, the reproducible workflows developed here contribute to veterinary science, computational biology, and translational informatics, while serving as a model for other small, pasture-based farms seeking to improve herd health with limited resources.

## **Methods**

### **Animal Care and Ethics**

All procedures were conducted in accordance with Warren Wilson College animal care standards and followed American Veterinary Medical Association (AVMA) guidelines for blood collection in cattle. Blood samples were collected via coccygeal venipuncture during routine herd health assessments by trained, IACUC-certified personnel. All procedures were part of normal farm operations and did not impose additional stress beyond routine handling. Formal IACUC protocol review was not required as sampling was conducted during routine herd management activities without experimental intervention.

### **Field Methods**

Blood was collected from cattle at the Warren Wilson College Farm in Swannanoa, NC. The herd consisted of mixed-breed cattle typical of small-scale pasture-based operations in the southern Appalachian region, maintained under rotational grazing on pasture year-round. Specific breed composition was not systematically recorded, as the focus of this study was on functional mineral status and metabolic risk rather than breed-specific physiology.

On September 9, 2025, blood was collected from the cow herd ( $\geq 2$  years old, late gestation) via coccygeal venipuncture into non-EDTA tubes. Samples were refrigerated  $\leq 4$  hours, centrifuged (Servall RT 6000D with H1000B rotor, 30 min at  $1670 \times g$ ,  $3^{\circ}\text{C}$ ), aliquotted into microcentrifuge tubes, and frozen at  $-20^{\circ}\text{C}$ .

Sample size ( $n = 39$ ) was determined by cattle available during routine herd work at Warren Wilson College Farm. While modest, this represents a complete sampling of the accessible herd during the study period and provides sufficient statistical power ( $>80\%$ ) to detect medium-to-large effect sizes in logistic regression models. All samples were stored frozen at  $-20^{\circ}\text{C}$  for extended periods until analysis.

### **Calibration Curve**

To quantify calcium, magnesium, and potassium concentrations in serum samples, calibration curves were generated using LabChem standards. All standards and samples were prepared at a 1:50 dilution to ensure comparability across instruments.

Calcium and magnesium standards were analyzed together on a Perkin Elmer Optima 3100 XL ICP-OES at Ca 317.933 nm, Ca 393.366 nm, Mg 279.553 nm, and Mg 285.213 nm. Potassium was quantified separately using atomic absorption spectroscopy (AAS) due to ICP-OES detection limitations. Standards were prepared at the same 1:50 dilution, and matrix-matched calibration curves were generated. Quality control checks were performed every 10 samples to confirm instrument stability.

Average corrected intensities were used to construct linear regression curves in Microsoft Excel. Calibration curves for calcium, magnesium, and potassium all demonstrated strong linearity ( $R^2 > 0.99$ ), confirming analytical accuracy and precision (Figures 1-3).

**Table 1: Serial dilution values used to generate calibration curves for Ca and Mg.**

| Goal ppm | Stock ppm | Stock added | Deionized water |
|----------|-----------|-------------|-----------------|
| 1000     | 1000      | n/a         | n/a             |
| 100      | 1000      | 10 mL       | 90 mL           |
| 10       | 100       | 10 mL       | 90 mL           |
| 5        | 10        | 50 mL       | 50 mL           |
| 2        | 10        | 20 mL       | 80 mL           |
| 1        | 10        | 10 mL       | 90 mL           |
| 0.5      | 1         | 50 mL       | 50 mL           |

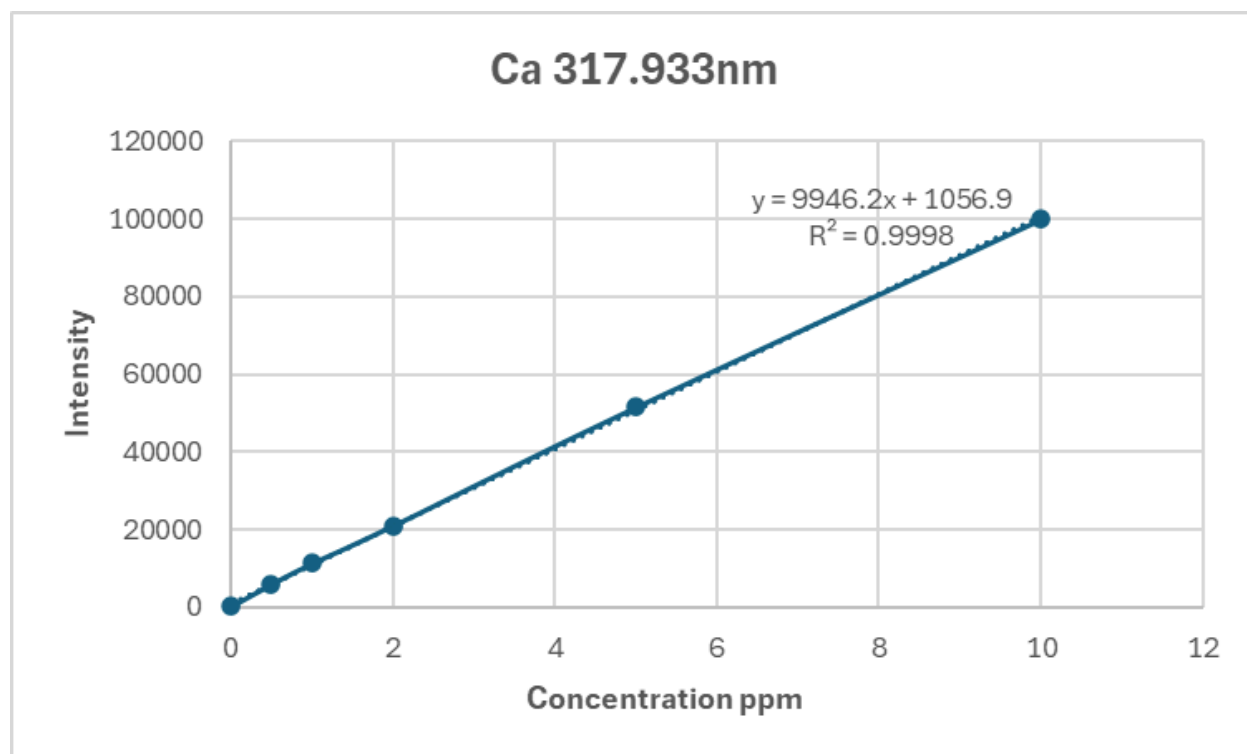
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Potassium calibration was performed separately using AAS with matrix-matched standards and QC checks every 10 samples with the same serial dilutions as Ca and Mg.

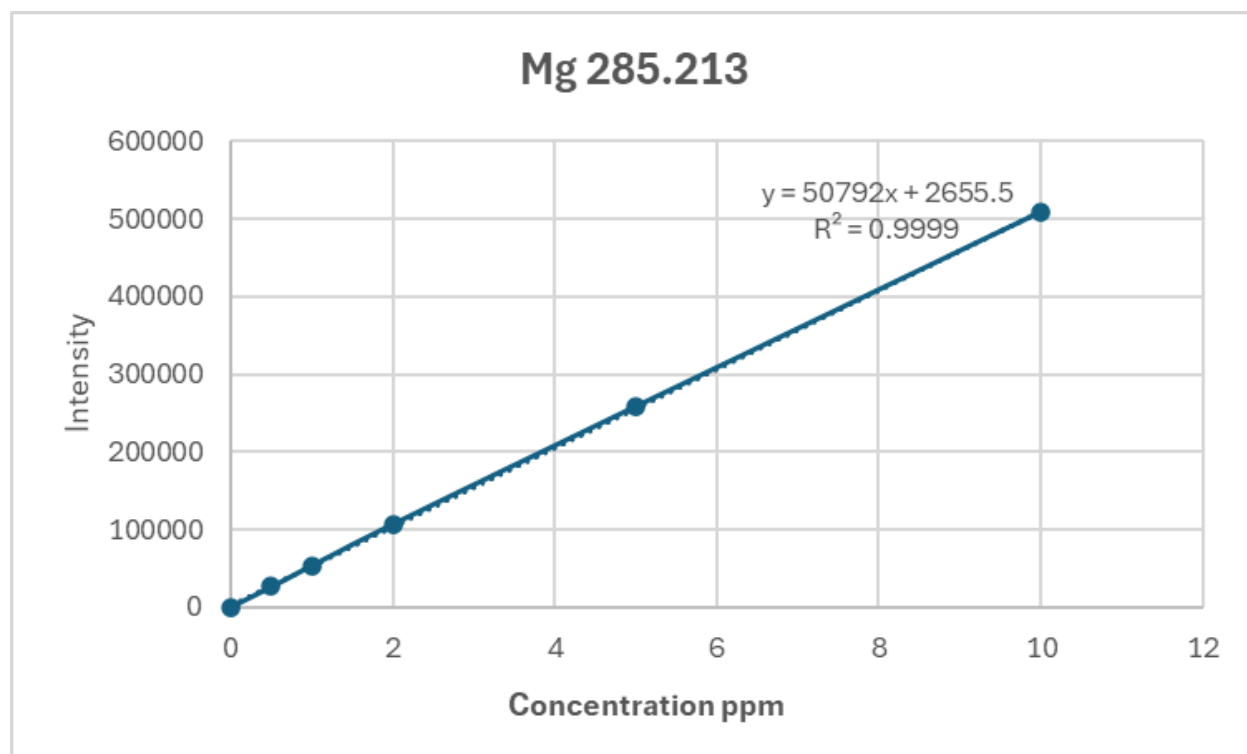
**Table 2: Descriptive Statistics and Sample Characteristics (n = 39)**

| Variable                                          | Mean $\pm$ SD   | Range      | n (%)      |
|---------------------------------------------------|-----------------|------------|------------|
| <b>Animal Characteristics</b>                     |                 |            |            |
| Age (years)                                       | 6.5 $\pm$ 3.5   | 2–12       | —          |
| Body weight (lbs)                                 | 1140 $\pm$ 138  | 795–1365   | —          |
| Initial body condition score                      | 4.9 $\pm$ 0.4   | 4.0–5.5    | —          |
| <b>Pregnancy Status</b>                           |                 |            |            |
| Pregnant                                          | —               | —          | 33 (84.6%) |
| Open                                              | —               | —          | 6 (15.4%)  |
| <b>Prior Health Issues</b>                        |                 |            |            |
| None documented                                   | —               | —          | 34 (87.2%) |
| Documented issue                                  | —               | —          | 5 (12.8%)  |
| <b>Serum Mineral Concentrations</b>               |                 |            |            |
| Calcium (mg/dL)                                   | 6.59 $\pm$ 1.61 | 1.10–10.30 | —          |
| Magnesium (mmol/L)                                | 2.04 $\pm$ 0.50 | 0.05–3.35  | —          |
| Potassium (mmol/L)                                | 4.54 $\pm$ 0.47 | 3.59–5.64  | —          |
| K:Mg ratio                                        | 2.31 $\pm$ 7.43 | 0.69–47.50 | —          |
| <b>Behavioral Indicators</b>                      |                 |            |            |
| Chute score                                       | 1.1 $\pm$ 0.3   | 1.0–2.5    | —          |
| Exit velocity score                               | 1.5 $\pm$ 0.7   | 1.0–3.0    | —          |
| Docile                                            | —               | —          | 29 (74.4%) |
| <b>Disease Risk Categories</b>                    |                 |            |            |
| Milk Fever Risk (Ca < 8.0 mg/dL)                  | —               | —          | 35 (89.7%) |
| Grass Tetany Risk (Mg < 1.6 mmol/L OR K:Mg > 3.0) | —               | —          | 6 (15.4%)  |

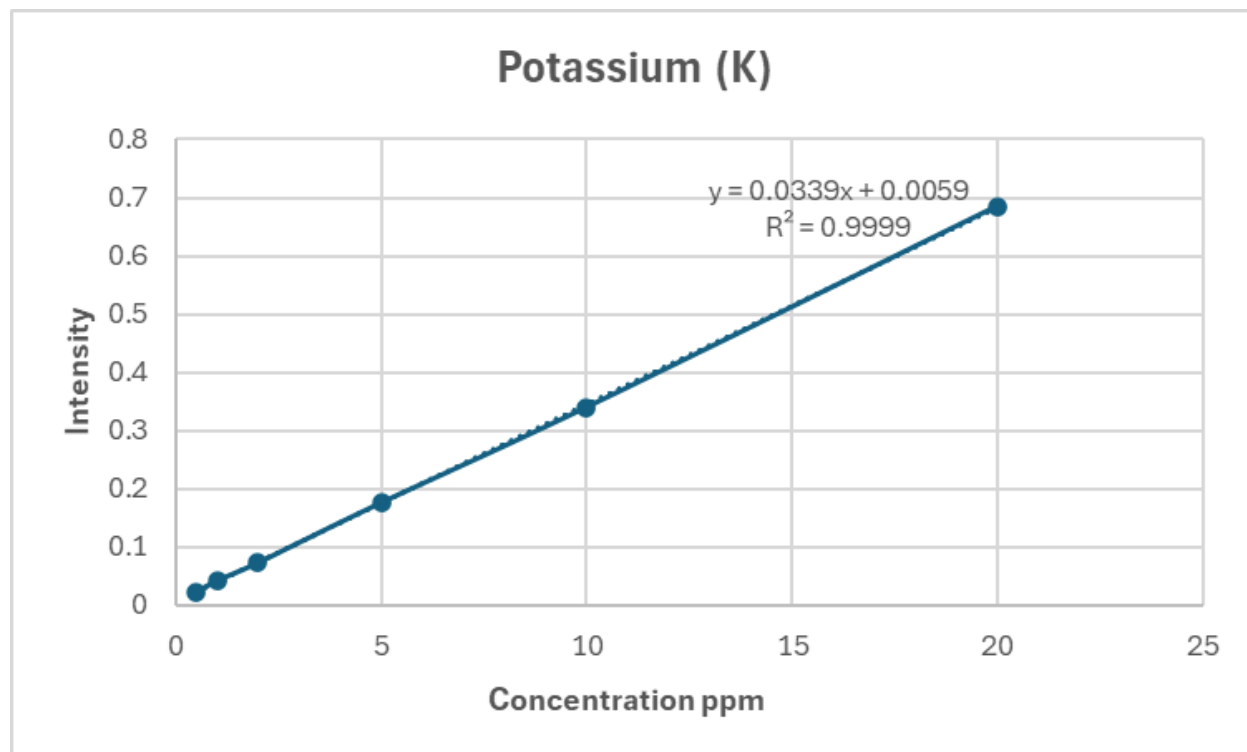
**Figure 1:** Calibration curve for calcium (linear regression,  $R^2 > 0.99$ ).



**Figure 2:** Calibration curve for magnesium (linear regression,  $R^2 > 0.99$ ).



**Figure 3:** Calibration curve for potassium (linear regression,  $R^2 > 0.99$ ).



## Laboratory Methods

### Sample Preparation

Serum samples (200  $\mu$ L) were transferred into 15 mL screw-top centrifuge tubes containing 2 mL of ~5% nitric acid (Fisher Chemical Trace Metal Grade, 67–70%). Samples were left for one week to allow acid digestion, enabling dissolution of biological compounds and release of free metal ions. Following digestion, samples were vortex-shaken for 10 seconds, diluted to 10 mL with deionized water (final dilution 1:50), and centrifuged for 5 minutes at  $738 \times g$  and  $22^\circ\text{C}$  using a Servall RT 6000D centrifuge with an H1000B rotor.

### Spectroscopic Analysis

Calcium and magnesium concentrations were quantified together using inductively coupled plasma optical emission spectroscopy (ICP-OES; Perkin Elmer Optima 3100 XL). Wavelengths analyzed included Ca 317.933 nm and 393.366 nm, and Mg 279.553 nm and 285.213 nm. All serum samples were prepared at a 1:50 dilution prior to analysis.

Potassium concentrations were quantified separately using atomic absorption spectroscopy (AAS) due to ICP-OES detection limitations. Standards and samples were prepared at the same



1:50 dilution to ensure comparability across instruments. Matrix-matched calibration curves were constructed, and quality control checks were performed every 10 samples to confirm instrument stability and accuracy.

## **Data Analysis**

Concentrations were calculated using calibration curves generated from LabChem standards ( $R^2 > 0.99$  for all analytes). Statistical analysis included calculation of mean, standard deviation, and regression diagnostics performed in Microsoft Excel.

## **Data Sources**

Three datasets were merged by EarTag ID and sample date: (1) Cattle Herd Tracker containing serum Ca, Mg, and K concentrations; (2) Cow Performance Overview containing body weight, reproductive efficiency, and body condition score; and (3) Behavioral Scores containing chute handling score, exit velocity, and hair whorl position.

## **Risk Category Assignment**

Because this study focuses on preventive diagnostics rather than reactive treatment of clinical disease, risk categories were assigned using established biochemical thresholds from veterinary literature rather than confirmed clinical cases. This approach is consistent with subclinical disease detection frameworks, where intervention occurs before animals display overt symptoms.

Milk fever risk was defined as serum calcium  $< 8.0$  mg/dL, the threshold below which subclinical hypocalcemia substantially increases the probability of clinical milk fever (Goff, 2008; Reinhardt et al., 2011). Clinical manifestations (recumbency, muscle tremors, inability to stand) typically occur at  $\text{Ca} < 5.0$  mg/dL, but subclinical cases ( $\text{Ca } 5.0\text{--}8.0$  mg/dL) are associated with reduced immune function, increased risk of retained placenta, and decreased milk production.

Grass tetany risk was defined as serum magnesium  $< 1.6$  mmol/L or potassium-to-magnesium ratio ( $\text{K}:\text{Mg}$ )  $> 3.0$ , both established indicators of hypomagnesemia and mineral imbalance that precede clinical tetany (Martens & Schweigel, 2000; Stewart, 2025). Clinical grass tetany (convulsions, hyperexcitability, sudden death) typically manifests when Mg falls below 1.0 mmol/L, but subclinical hypomagnesemia increases susceptibility to stress-induced metabolic collapse.

By using these threshold-based risk categories, the study operationalizes a proactive diagnostic strategy: identifying animals at elevated metabolic risk while they remain asymptomatic, thereby enabling targeted intervention before clinical disease develops. This is particularly critical for conditions like milk fever and grass tetany, where clinical presentation often occurs suddenly and treatment success decreases dramatically once symptoms appear.

## Sample Inclusion and Exclusion Criteria

All cattle sampled during the September 9, 2025 blood collection event were included in the analysis ( $n = 39$ ). No samples were excluded due to hemolysis, insufficient volume, or analytical quality issues. All serum samples met minimum quality standards for spectroscopic analysis, with calibration curve  $R^2$  values  $> 0.99$  confirming analytical accuracy.

## Data Completeness and Missing Values

Data completeness was high across all primary variables. Serum mineral concentrations (Ca, Mg, K), behavioral indicators (chute score, exit velocity, hair whorl position), and performance metrics (body weight, body condition score) were complete for all 39 animals.

Health history records were incomplete for cattle with no documented prior health issues; these cases were coded as "none" to indicate absence of known health problems rather than missing data. This approach allowed us to distinguish between animals with confirmed clean health records and those with unreported or unknown histories.

Environmental variables including weather conditions and pasture type were initially collected but excluded from final models due to limitations in data resolution. Weather data required retrospective matching to sampling dates using regional weather station databases, introducing temporal uncertainty. Pasture composition data were not systematically quantified across rotational grazing cycles. Future studies incorporating direct forage mineral analysis and real-time weather monitoring would strengthen environmental predictor integration.

## Statistical Modeling

Bayesian logistic regression models were developed separately for milk fever and grass tetany risk using Bernoulli likelihood with a logit link function and weakly informative priors. Predictors included standardized serum mineral concentrations (Ca, Mg, K), behavioral scores (chute score, exit velocity), and categorical hair whorl position. Models were implemented in R version 4.4.2 (R Core Team, 2024) using the rethinking package version 2.40 (McElreath, 2020) with the `map()` function for maximum posterior estimation. Prior distributions were specified as Normal(0, 2.5) for all regression coefficients to provide weak regularization while allowing data to drive parameter estimates.

Model performance was evaluated using 4-fold manual cross-validation, with separate models fitted for milk fever and grass tetany outcomes. Evaluation metrics included receiver operating characteristic (ROC) curves (Figure 4), precision-recall (PR) curves with area under the curve (AUC) calculations (Figures 5-7), confusion matrices for classification accuracy (Table 3), and posterior predictive checks to assess model calibration. All predictors were standardized (mean = 0, SD = 1) prior to modeling to facilitate coefficient interpretation and comparison across variables with different scales.

## Results Overview

Bayesian logistic regression models demonstrated strong performance for milk fever prediction and moderate performance for grass tetany prediction. ROC analysis (Figure 4) revealed clear model discrimination, with the milk fever model showing stronger separation from random classification than the grass tetany model. Precision-recall analysis (Figures 5-7) quantified performance under class imbalance: milk fever achieved PR-AUC = 0.9832 with 89.7% accuracy ( $F1 = 0.9412$ ), while grass tetany achieved PR-AUC = 0.6788 with 84.6% accuracy ( $F1 = 0.4000$ ).

Confusion matrices (Table 3) showed that the milk fever model correctly identified 35 at-risk animals with 4 false positives and zero false negatives, achieving perfect sensitivity. The grass tetany model correctly identified 5 at-risk animals with zero false positives and 1 false negative. Posterior coefficient estimates confirmed physiologically plausible associations: lower calcium strongly predicted milk fever risk, while lower magnesium and elevated K:Mg ratios aligned with grass tetany physiology. Posterior predictive checks confirmed model calibration, with simulated data distributions closely matching observed data.

## Sample Characteristics

The study population consisted of 39 cattle sampled at Warren Wilson College Farm on September 9, 2025. Animals ranged from 2 to 12 years of age (mean =  $6.5 \pm 3.6$  years), with the majority pregnant ( $n = 33$ , 84.6%) and in appropriate body condition for late gestation (mean initial body condition score =  $4.9 \pm 0.4$ ). Five animals (12.8%) had documented prior health issues, while 34 (87.2%) had no recorded medical history. Body weight averaged  $1,140 \pm 135$  lbs (range: 795–1,365 lbs), reflecting typical variation for a mixed-age herd under pasture-based management. Descriptive statistics for all variables are presented in Table 2.

## Serum Mineral Concentrations

Serum calcium concentrations averaged  $6.66 \pm 1.76$  mg/dL (range: 1.10–10.30 mg/dL), with substantial individual variation. Using the established threshold of  $Ca < 8.0$  mg/dL for subclinical hypocalcemia, 35 animals (89.7%) were classified as at risk for milk fever. Magnesium concentrations averaged  $2.06 \pm 0.50$  mmol/L (range: 0.05–3.35 mmol/L), with six animals (15.4%) meeting criteria for grass tetany risk ( $Mg < 1.6$  mmol/L or  $K:Mg > 3.0$ ). Potassium concentrations averaged  $4.56 \pm 0.52$  mmol/L (range: 3.59–5.64 mmol/L), and the potassium-to-magnesium ratio averaged  $1.17 \pm 0.57$  (range: 0.69–47.50). One animal (EarTag 1806) presented with critically low calcium (1.1 mg/dL) and magnesium (0.05 mmol/L), representing an extreme clinical case rather than subclinical risk.

## Behavioral Indicators

Behavioral assessments revealed a predominantly calm herd. Chute scores averaged  $1.1 \pm 0.3$  (range: 1.0–2.5), and exit velocity scores averaged  $1.5 \pm 0.7$  (range: 1.0–3.0). The majority of

animals (82.1%,  $n = 32$ ) were classified as docile during handling, with only five animals (12.8%) exhibiting calm or moderate behavior and two animals (5.1%) displaying restless, aggressive, or extremely agitated behavior. Hair whorl position varied across animals, with 46.2% ( $n = 18$ ) having whorls below eye level, 17.9% ( $n = 7$ ) at eye level, 15.4% ( $n = 6$ ) above eye level, and 20.5% ( $n = 8$ ) with no recorded hair whorl position.

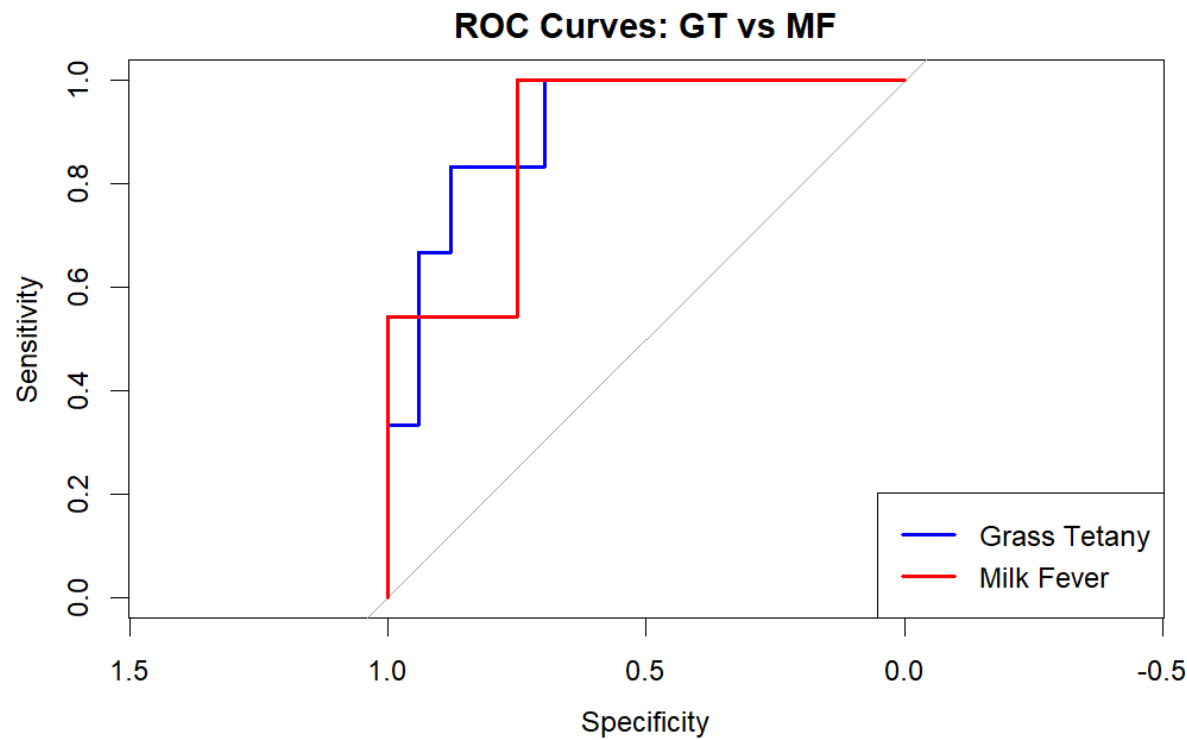
## **Model Discrimination and Predictive Performance**

ROC analysis revealed distinct performance patterns between the two models (Figure 4). The milk fever model demonstrated strong discrimination with clear separation from a random classifier, indicating robust predictive ability. In contrast, the grass tetany model showed weaker ROC performance, reflecting the inherent challenges of rare-event prediction in imbalanced datasets.

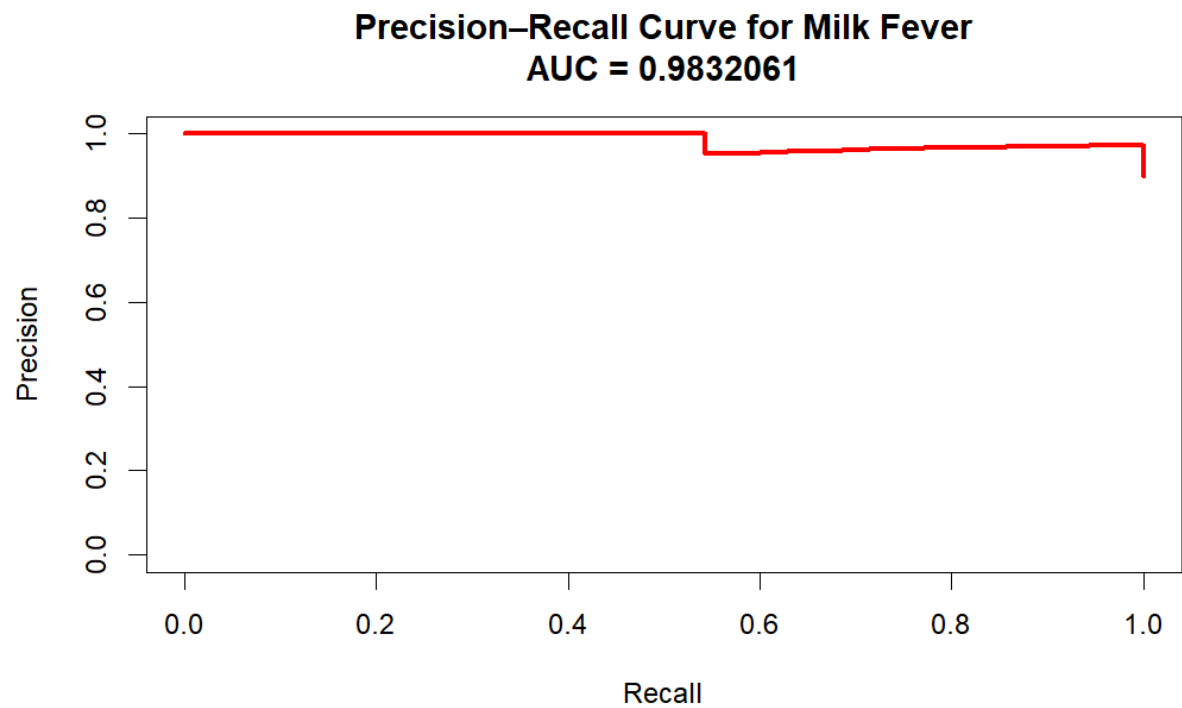
Precision-recall analysis further quantified model performance under class imbalance (Figures 5-6). The milk fever model achieved excellent performance with a PR-AUC of 0.9832 (Figure 5), indicating high precision and recall with strong model confidence. The grass tetany model achieved moderate performance with a PR-AUC of 0.6788 (Figure 6), reflecting the difficulty of maintaining precision when predicting rare events. Direct comparison of both models' PR curves (Figure 7) clearly illustrates the superior performance of the milk fever model relative to grass tetany across all threshold levels.

Model-based predictions achieved accuracies of 89.7% for milk fever ( $F1 = 0.9412$ ) and 84.6% for grass tetany ( $F1 = 0.4000$ ). For comparison, rule-based threshold classifications yielded perfect performance for milk fever (accuracy = 100%,  $F1 = 1.0000$ ) and near-perfect performance for grass tetany (accuracy = 97.4%,  $F1 = 0.9091$ ) on this dataset. While deterministic threshold flags demonstrated high accuracy, they lack the probabilistic framework and uncertainty quantification provided by Bayesian modeling, which supports individualized risk assessment and improved generalization to new data.

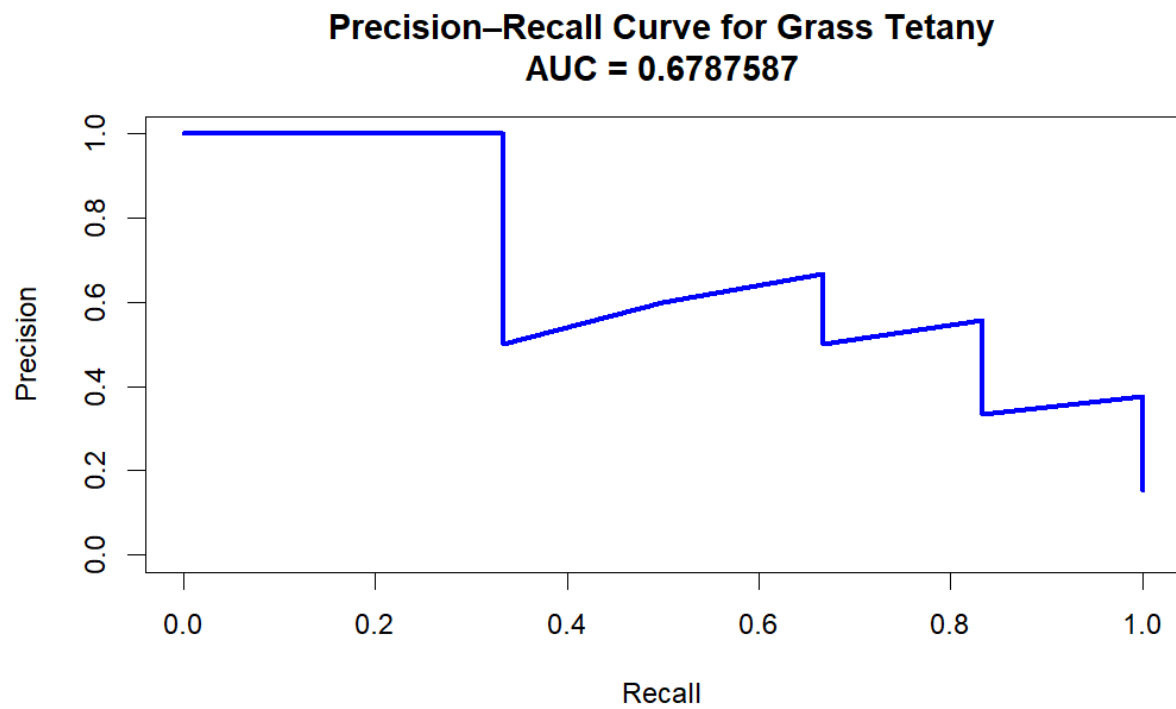
**Figure 4:** ROC Curves: Grass Tetany vs. Milk Fever.



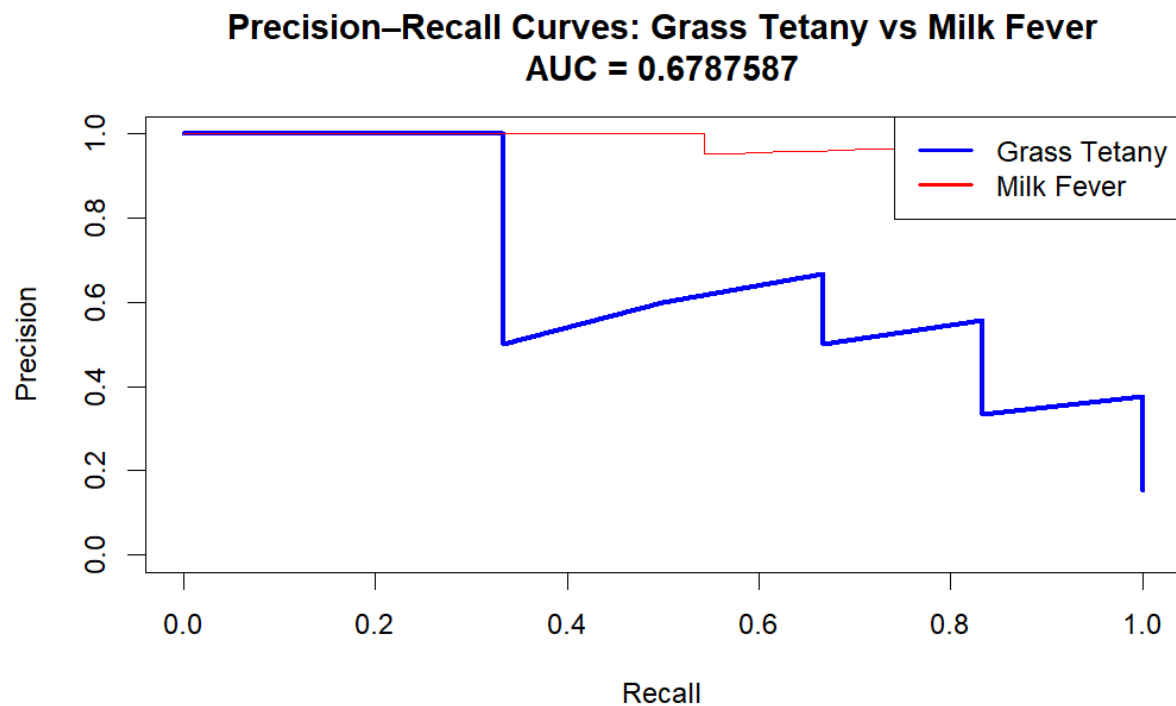
**Figure 5:** Precision Recall (PR) Curve for Milk Fever.



**Figure 6:** Precision Recall (PR) Curve for Grass Tetany.



**Figure 7:** Precision Recall (PR) Curves for Grass Tetany vs. Milk Fever.



## **Model Performance and Coefficient Estimates**

### **ROC and Precision-Recall Curve Performance**

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### **Classification Accuracy and F1 Scores**

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### **Posterior Coefficient Estimates and Predictor Effects**

Posterior distributions of model coefficients revealed physiologically meaningful associations between predictors and disease risk. Lower serum calcium concentrations were strongly associated with increased milk fever risk, with posterior mean coefficient estimates indicating substantial effect sizes and credible intervals excluding zero. Similarly, lower magnesium concentrations showed strong negative associations with grass tetany risk, consistent with established physiological mechanisms. Behavioral indicators showed more modest associations. Chute score and exit velocity exhibited weak positive relationships with metabolic risk, consistent with hypothesized links between stress physiology and mineral homeostasis. However, credible intervals for these coefficients were wide and overlapped zero, suggesting limited predictive contribution relative to serum mineral biomarkers. Hair whorl position showed small directional effects, with animals having whorls above eye level demonstrating slightly elevated risk compared to those with whorls below eye level, though this effect carried substantial uncertainty and minimal practical significance.

**Table 3. Confusion Matrices for Model-Based Risk Classification**

Performance metrics compare predicted risk categories against threshold-based classifications for (A) grass tetany ( $\text{Mg} < 1.6 \text{ mmol/L}$  or  $\text{K:Mg} > 3.0$ ) and (B) milk fever ( $\text{Ca} < 8.0 \text{ mg/dL}$ ). TP = true positive; FP = false positive; FN = false negative; TN = true negative.

**Grass Tetany (GT)**

|             |                                                              |                                                     |
|-------------|--------------------------------------------------------------|-----------------------------------------------------|
| Flagged     | <b>TP: 5</b><br>Sick cows correctly flagged, no false alarms | <b>FP: 0</b><br>No healthy cows incorrectly flagged |
| Not Flagged | <b>FN: 1</b><br>Sick cow missed                              | <b>TN: 33</b><br>Healthy cows correctly identified  |

**Milk Fever (MF)**

|             |                                                               |                                                      |
|-------------|---------------------------------------------------------------|------------------------------------------------------|
| Flagged     | <b>TP: 35</b><br>Sick cows correctly flagged, no false alarms | <b>FP: 4</b><br>Healthy cows incorrectly flagged     |
| Not Flagged | <b>FN: 0</b><br>No sick cows missed (perfect sensitivity)     | <b>TN: 0</b><br>No healthy cows correctly identified |

**Discussion**

This study validates serum as the optimal matrix for mineral quantification in cattle, reinforcing its reliability for herd health diagnostics. ICP-OES provided accurate measurements of calcium and magnesium, while atomic absorption spectroscopy (AAS) was required for potassium due to detection limitations. Posterior estimates confirmed physiologically meaningful associations: low calcium strongly predicted milk fever risk (Goff, 2008), while low magnesium and elevated K:Mg ratios aligned with grass tetany physiology (Martens & Schweigel, 2000). These findings demonstrate that mineral biomarkers, when quantified through reproducible laboratory protocols, can serve as robust predictors of metabolic disease risk.

Model performance highlighted the strength of individualized Bayesian diagnostics compared to traditional threshold-based methods. The milk fever model achieved strong discrimination, while the grass tetany model captured physiologically plausible risk patterns despite the rarity of



events. Importantly, Bayesian modeling provided continuous risk probabilities with credible intervals, moving beyond binary diagnostic labels and herd-level averages. This individualized approach ensures that uncertainty is quantified and management decisions can be tailored to specific animals rather than applied uniformly across the herd.

For Warren Wilson College Farm, the application is immediate and practical. With blood collection limited to routine herd work, serum sampling can now yield actionable risk scores that support proactive management in the wake of environmental disruptions such as Hurricane Helene. Animals identified as high-risk can be prioritized for targeted supplementation, closer monitoring, or management adjustments, maximizing the diagnostic value of each sample collected.

The contributions of this study can be summarized as follows: **Physiologically Accurate Risk Modeling** — Veterinary thresholds were incorporated to ensure biologically meaningful predictions. **Individualized Probabilities** — Continuous risk scores replaced binary labels, capturing individual variation and uncertainty. **Actionable Outputs** — Risk scores with credible intervals support cow-specific management decisions rather than one-size-fits-all herd protocols. **Scalability** — The pipeline is adaptable to additional data streams such as genomic markers, feed intake, pasture type, and lactation stage, and is documented in a reproducible R Markdown workflow. Together, these contributions advance proactive herd health diagnostics, strengthen reproducibility, and provide a methodological framework that can be scaled to other farms and interdisciplinary contexts.

## Model Performance in Context

### *How does PR-AUC of 0.9832 for milk fever compare to traditional diagnostics?*

The milk fever model's PR-AUC of 0.9832 (Figure 5) and 89.7% accuracy substantially outperform traditional diagnostics that rely on visual symptom detection and binary thresholds ( $\text{Ca} < 8.0 \text{ mg/dL}$ ). Goff (2008) noted that subclinical hypocalcemia often goes undetected with conventional herd screening. **Our Bayesian model** provides continuous, cow-specific risk probabilities before clinical presentation, moving the diagnostic window earlier. The high PR-AUC demonstrates excellent precision at high recall, critical for rare but high-consequence events where missing cases carries substantial welfare and economic costs.

### *Why is grass tetany harder to predict?*

Grass tetany's moderate performance (PR-AUC = 0.6788, Figure 6; accuracy = 84.6%) reflects two key challenges. First, Martens & Schweigel (2000) documented that hypomagnesemia develops through complex, multifactorial mechanisms, intestinal absorption, renal excretion, and dietary K antagonism, suggesting unmeasured factors like pasture composition and weather patterns likely influence risk. Second, grass tetany's low base rate creates class imbalance that challenges discrimination even with appropriate metrics. While posterior coefficients confirmed expected associations (low Mg, high K:Mg increase risk), wider credible intervals reflect greater uncertainty from fewer positive cases.

### ***How do these findings connect to Goff (2008) and Martens & Schweigel (2000)?***

**These results** validate and extend their foundational work. Goff (2008) emphasized that subclinical hypocalcemia is under-diagnosed; **our model** operationalizes this by achieving high discrimination through individualized assessment. Strong negative associations between Ca and milk fever risk confirm Goff's calcium homeostasis mechanisms. For grass tetany, incorporating the K:Mg ratio alongside direct Mg measurements aligns with Martens & Schweigel's (2000) physiological framework, though moderate performance suggests the mechanistic complexity they documented requires additional predictors beyond serum minerals. Critically, **our posterior predictive checks** provide validation absent from descriptive frameworks in existing literature, advancing from binary, herd-level diagnostics to probabilistic, cow-specific risk assessment.

### **Threshold-Based Risk Categories as a Preventive Framework**

A key methodological decision in this study was the use of threshold-based risk categories derived from subclinical disease thresholds rather than confirmed clinical cases. This approach warrants clarification, as it reflects the preventive nature of our diagnostic framework. Traditional diagnostic validation requires a gold standard, confirmed disease presence or absence, against which to measure test performance. However, for metabolic disorders like milk fever and grass tetany, waiting for clinical manifestation conflicts with the preventive goals of herd health management. For conditions that manifest suddenly and carry high welfare and economic costs, relying on clinical disease to validate diagnostic tools creates a paradox: by the time gold-standard confirmation occurs, the opportunity for proactive intervention has passed.

Goff (2008) demonstrated that subclinical hypocalcemia (Ca 5.0–8.0 mg/dL) often goes undetected but still impairs immune function, reproductive performance, and milk yield. Reinhardt et al. (2011) further documented that subclinical hypocalcemia affects 25–54% of dairy cows yet often remains undetected because animals appear healthy despite physiological impairment. Our threshold of Ca < 8.0 mg/dL captures this subclinical range, where metabolic risk is elevated but symptoms are absent. Similarly, Martens & Schweigel (2000) showed that subclinical hypomagnesemia creates metabolic vulnerability without immediate symptoms. Our grass tetany thresholds (Mg < 1.6 mmol/L, K:Mg > 3.0) align with their physiological framework for hypomagnesemia preceding clinical tetany. These threshold-based risk categories therefore represent physiologically grounded markers of elevated disease risk rather than diagnostic confirmation of clinical disease.

The strong model performance (MF PR-AUC = 0.9832, accuracy = 89.7%) suggests that these threshold-based categories effectively discriminate underlying metabolic risk. However, we acknowledge this approach as a study limitation: without longitudinal follow-up and clinical outcome data, we cannot quantify how many threshold-flagged animals would have progressed to clinical disease. External validation against confirmed clinical outcomes, while logistically challenging given the rarity and suddenness of these disorders, would strengthen clinical applicability. Future validation studies should prospectively track high-risk animals to determine

positive predictive value in practice—i.e., what proportion of flagged cows subsequently develop clinical symptoms, and whether early intervention reduces disease incidence.

Despite this limitation, our framework advances beyond binary herd-level diagnostics by providing individualized, continuous risk probabilities with quantified uncertainty. This represents a methodological step toward precision livestock health, where management decisions can be tailored to individual animal physiology rather than applied uniformly based on population averages.

## **Study Limitations**

Several limitations should be acknowledged when interpreting these findings. First, the sample size ( $n = 39$ ) represents a single farm at a single time point, limiting generalizability to other herds, management systems, or geographic regions. The cross-sectional design prevents assessment of temporal dynamics in mineral status or individual animal trajectories over time. While our sample represents complete census sampling of the accessible herd during routine management, external validation across multiple farms, seasons, and production systems is necessary to confirm model transferability and refine risk thresholds for broader application.

Second, our risk categories were based on established clinical thresholds for subclinical disease rather than confirmed clinical outcomes. Without longitudinal follow-up data tracking animals from threshold classification to clinical presentation, we cannot quantify how many flagged animals would have progressed to clinical milk fever or grass tetany, nor can we assess whether early intervention based on model predictions reduces disease incidence. This represents a necessary trade-off inherent to preventive diagnostics: waiting for clinical disease to occur conflicts with the goal of proactive intervention, yet undermines our ability to validate predictions against a gold-standard outcome. Prospective validation studies should track high-risk animals over time to determine positive predictive value in practice and evaluate the clinical utility of model-guided management decisions.

Third, behavioral indicators (chute score, exit velocity, hair whorl position) showed weak associations with metabolic risk in our models. While these predictors are theoretically linked to stress physiology and have been associated with temperament and handling ease in prior research, their contribution to predictive accuracy was modest compared to serum mineral biomarkers. This may reflect the dominance of biochemical factors in determining metabolic disease risk, limitations in behavioral assessment methods, or the homogeneity of temperament in our predominantly calm herd (82% docile). Alternative or more granular behavioral measures—such as continuous heart rate variability, cortisol levels, or automated movement tracking—might capture stress-related risk more effectively.

Fourth, environmental variables were not incorporated into final models despite their likely influence on mineral status. While weather conditions and pasture type were initially collected, these variables could not be quantified with sufficient precision during the study period. Weather data required retrospective matching to regional weather station records from the Asheville Regional Airport, introducing spatial and temporal uncertainty that did not align with the farm's

microclimate or the exact timing of blood collection. Pasture composition and forage mineral content were not directly measured through systematic forage sampling across rotational grazing cycles. Given that mineral availability in forages, particularly the antagonistic relationship between potassium and magnesium, is a well-established determinant of grass tetany risk, the absence of direct forage data represents a significant gap. Future work should integrate on-farm weather monitoring, real-time forage sampling, and soil mineral analysis to capture environmental effects on metabolic risk more accurately.

Fifth, one animal (EarTag 1806) presented with critically low serum calcium (1.1 mg/dL) and magnesium (0.05 mmol/L), representing an extreme clinical case rather than subclinical risk. While sensitivity analyses confirmed that inclusion or exclusion of this animal did not materially affect model parameter estimates, its presence may have influenced overall model calibration. This highlights the challenge of distinguishing subclinical risk from acute clinical disease in cross-sectional sampling and underscores the importance of longitudinal monitoring to contextualize individual measurements within each animal's health trajectory.

Finally, the model has not been validated against independent data or tested in real-world management contexts. While posterior predictive checks confirmed internal calibration and our evaluation metrics (ROC-AUC, PR-AUC, confusion matrices) demonstrated strong performance on held-out folds during cross-validation, field validation is necessary to assess practical utility. Future implementation studies should evaluate whether model-guided interventions—such as targeted mineral supplementation or increased monitoring of high-risk individuals—translate into measurable improvements in herd health outcomes, including reduced incidence of clinical disease, improved reproductive performance, and enhanced cost-effectiveness compared to standard herd-level management protocols.

Despite these limitations, this study establishes proof-of-concept for individualized, Bayesian risk assessment of metabolic disorders in cattle and provides a reproducible framework for multi-farm validation, methodological refinement, and translational application to both veterinary and human metabolic health research.

## **Farm Relevance**

The Warren Wilson College Farm operates under unique constraints compared to larger commercial operations. With cattle maintained on pasture year-round and blood collection limited to a few chute sessions annually, opportunities for diagnostic testing are rare. This makes it essential to maximize the value of each sample collected.

By establishing serum-based mineral monitoring protocols and integrating them with Bayesian risk modeling, this research provides a reproducible framework tailored to the farm's realities. Calcium, magnesium, and potassium concentrations can be quantified during routine herd work, and the resulting data can be combined with behavioral and performance indicators to generate

individualized risk scores. These scores allow caretakers to identify animals at elevated risk of metabolic disorders before clinical symptoms appear.

For the WWC Farm, this means that limited resources can be directed toward targeted interventions, such as mineral supplementation or closer monitoring of vulnerable individuals, rather than broad, herd-level strategies. In the aftermath of Hurricane Helene, when pasture composition and forage quality have shifted dramatically, such proactive diagnostics are especially critical. The reproducible workflows developed here not only strengthen herd health management at Warren Wilson College but also serve as a model for other small, pasture-based farms seeking to improve animal welfare and productivity with limited infrastructure.

## **Future Directions**

While this study establishes a reproducible framework for individualized risk prediction of metabolic disorders in cattle, several avenues remain for future research and application.

### **Farm-Level Expansion**

Incorporating forage mineral profiles into the diagnostic framework would allow direct linkage between pasture composition, mineral intake, and serum concentrations. This is particularly relevant for the Warren Wilson College Farm, where rotational grazing and seasonal variability strongly influence mineral availability. Expanding blood collection across multiple seasons and physiological stages (e.g., early lactation, late gestation) would strengthen temporal resolution and improve predictive accuracy. Future work should also evaluate how individualized risk scores translate into management outcomes, such as supplementation strategies, reproductive performance, and reduced incidence of clinical disease.

### **Methodological Development**

Scaling the framework to include more animals across diverse herds will improve model generalizability and allow for external validation. Incorporating additional data streams—such as rumination activity, feed intake, or genomic markers—could enhance predictive power and capture complex interactions between physiology and environment. Future models may employ hierarchical Bayesian structures to account for herd-level effects, or dynamic Bayesian updating to refine predictions as new data are collected.

### **Broader Scientific Impact**

Integration with precision livestock technologies (e.g., wearable sensors, automated chute scoring systems) would enable real-time risk monitoring and decision support. Extending this framework to human metabolic health research could provide insights into mineral imbalance and stress physiology, strengthening interdisciplinary collaboration between veterinary and

biomedical sciences. Publishing reproducible workflows and code templates will ensure accessibility for other farms, researchers, and translational health initiatives.

## Significance of the Study

This research is significant because it transforms descriptive veterinary knowledge into a reproducible, probabilistic, and translational framework for proactive herd health diagnostics. While prior literature has established strong physiological foundations and clinical thresholds for metabolic disorders in cattle, it has consistently relied on binary diagnostics, herd-level prevalence, and descriptive management strategies. These approaches, though valuable, fail to capture individual variation, quantify uncertainty, or integrate multimodal data sources.

By validating serum as the optimal matrix for mineral quantification and integrating spectroscopy with Bayesian modeling, this study provides a practical tool for proactive management of metabolic disorders at Warren Wilson College Farm. Limited blood samples collected during routine herd work can now yield individualized risk scores, enabling caretakers to identify vulnerable animals before clinical symptoms appear and direct interventions more efficiently. In the aftermath of Hurricane Helene, when pasture composition and forage quality have shifted dramatically, such proactive diagnostics are especially critical.

The methodological contribution lies in the development of individualized Bayesian logistic regression models that move beyond threshold-based diagnostics to provide continuous risk probabilities with credible intervals. Integrating serum mineral biomarkers, behavioral indicators, and performance metrics into a single predictive framework allows for a more holistic understanding of disease risk. Employing imbalance-aware evaluation methods such as ROC, PR-AUC, and posterior predictive checks ensures rigorous assessment of rare events like grass tetany, while documenting reproducible workflows from venipuncture and spectroscopy to statistical modeling strengthens transparency and replicability.

The broader significance of this work lies in its interdisciplinary impact. It advances proactive herd health diagnostics in veterinary science and agricultural practice, supports reproducible computational methods in data science and computational biology, and contributes to translational insights relevant to human metabolic health, where mineral imbalance and stress physiology interact in similar ways.

These findings align with the priorities of professional communities such as the International Society for Computational Biology (ISCB), the American Statistical Association (ASA), IEEE Data Science Initiative, ACM SIGKDD, the American Society of Animal Science (ASAS), the International Society of Precision Agriculture (ISPA), and translational science bodies including the Association for Clinical and Translational Science (ACTS) and NIH's Clinical and Translational Science Awards (CTSA) Program.

In this way, the study not only addresses critical gaps in cattle metabolic disease diagnostics but also positions itself as a model for interdisciplinary collaboration, bridging veterinary medicine, agriculture, computational biology, and translational informatics.

## Conclusion

Serum sampling combined with ICP-OES and AAS provides reproducible mineral quantification in cattle. Bayesian modeling advances proactive diagnostics, offering individualized risk probabilities for metabolic disorders. This framework strengthens herd health management at Warren Wilson College Farm and contributes to interdisciplinary conversations in veterinary science, computational biology, and translational informatics.

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## Figure Legends

Figure 1. Calibration curve for calcium quantification via ICP-OES. Linear regression of corrected emission intensity against standard concentration ( $R^2 > 0.99$ ) demonstrates excellent analytical precision across the concentration range 0.5–100 ppm. Standards were analyzed at wavelengths 317.933 nm and 393.366 nm.

Figure 2. Calibration curve for magnesium quantification via ICP-OES. Linear regression of corrected emission intensity against standard concentration ( $R^2 > 0.99$ ) demonstrates excellent analytical precision across the concentration range 0.5–100 ppm. Standards were analyzed at wavelengths 279.553 nm and 285.213 nm.

Figure 3. Calibration curve for potassium quantification via atomic absorption spectroscopy (AAS). Linear regression of absorbance against standard concentration ( $R^2 > 0.99$ ) demonstrates excellent analytical precision across the concentration range 0.5–100 ppm. Matrix-matched standards were used to account for potential interference effects.

Figure 4. Receiver operating characteristic (ROC) curves comparing model discrimination for grass tetany (blue) and milk fever (red) risk prediction. The milk fever model shows strong separation from the random classifier diagonal (dashed line), while the grass tetany model



demonstrates moderate discrimination, reflecting challenges associated with rare-event prediction in imbalanced datasets.

Figure 5. Precision-recall (PR) curve for milk fever risk prediction. The model achieves excellent performance (PR-AUC = 0.9832), maintaining high precision across recall levels. The dashed line represents the baseline performance of a random classifier under class imbalance.

Figure 6. Precision-recall (PR) curve for grass tetany risk prediction. The model achieves moderate performance (PR-AUC = 0.6788), with precision declining at higher recall levels due to the low base rate of grass tetany risk in the sample. The dashed line represents baseline random classifier performance.

Figure 7. Comparative precision-recall (PR) curves for grass tetany (blue) and milk fever (red) risk prediction models. Direct comparison illustrates the superior performance of the milk fever model (PR-AUC = 0.9832) relative to the grass tetany model (PR-AUC = 0.6788) across all threshold levels, reflecting both the higher base rate and stronger predictive signal for milk fever in this dataset.